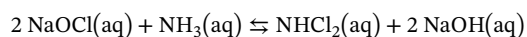


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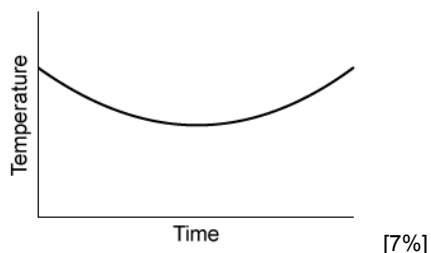
Table 1 Partial Data for Enthalpies, Entropies, and Free Energies of Formation of Several Compounds at 25 °C

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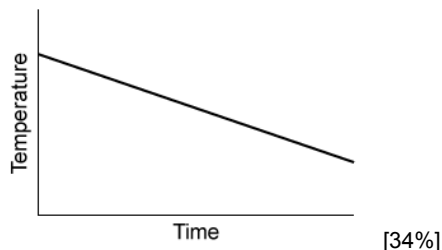
After it is formed, dichloramine readily enters the gas phase, where it may be inhaled. In the lungs, dichloramine reacts with water to produce hydrochloric acid and ammonia, as well as reactive oxygen species. In small doses, these products cause mild irritation, but inhalation of large amounts of chloramines can lead to pneumonitis (inflammation of the lungs) and may require medical intervention.

Which of the following graphs best depicts the expected change in temperature of the reaction mixture over the course of the reaction described in the passage?

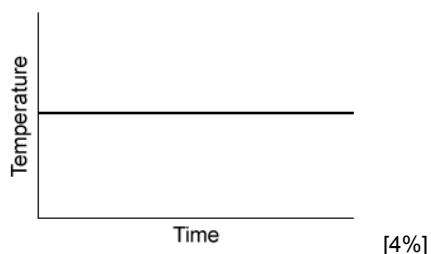
☐ A.



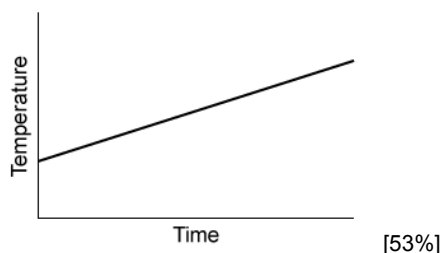
☐ B.



☐ C.



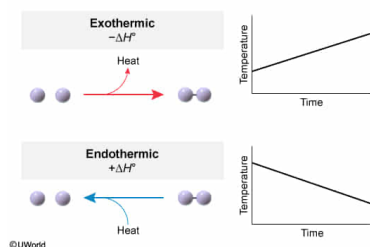
☒ D.



Correct

54% answered correctly

Explanation:



The heat released or consumed in a chemical process is called **enthalpy**, and the change in enthalpy is expressed by the term ΔH° . During chemical reactions, molecules convert their **internal (chemical) energy** into thermal energy (heat) as new bonds form, and they also convert thermal energy into internal energy as old bonds break.

If the heat produced by bond formation exceeds the heat consumed by bond cleavage, the overall reaction is **exothermic** and has a **negative value for ΔH°** . The net conversion of chemical energy to thermal energy causes the **temperature** of the system and its immediate surroundings to **increase**. Consider, for example, how the exothermic combustion of wood causes both the wood (system) and the air around it (surroundings) to heat up. Conversely, if more heat is consumed than produced, the reaction is **endothermic** with a **positive ΔH°** and the **temperature** of the system and its immediate surroundings **decreases**.

According to the passage, ΔH° for the overall reaction is negative (-55 kJ/mol), making the process exothermic. Therefore, heat is released as the reaction proceeds, and the temperature of the reaction mixture (ie, the system) will increase.

(Choice A) This graph shows an endothermic process (temperature decreases) followed by an exothermic process (temperature increase), and the final temperature is equal to the initial temperature (no net increase in temperature). The reaction in the passage is entirely exothermic (negative ΔH°), so the final temperature should be higher than the initial temperature.

(Choice B) The temperature of the system would decrease if it were simply **transferring existing** heat to its surroundings or if the reaction were **endothermic**. However, in an **exothermic reaction**, **new** heat is produced from chemical energy. This new heat increases the temperature of **both** the system and the surroundings.

(Choice C) The temperature of the reaction would remain constant only if ΔH° were equal to zero.

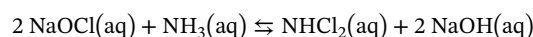
Educational objective:

The change in enthalpy of a reaction ΔH° determines whether heat is released or consumed. A negative value for ΔH° corresponds to an exothermic reaction in which heat is released and temperature rises. A positive value indicates an endothermic reaction and a decrease in temperature.

Time spent: 0 sec
Subject:
General Chemistry

QID: 401188
System:
Thermodynamics, Kinetics & Gas Laws

Household cleaners commonly contain either ammonia (NH₃) or bleach (NaOCl) as the principal active ingredient. Mixing ammonia-based and bleach-based cleaners can be hazardous, as the resulting reactions can form several toxic products. One of these reactions forms the irritant dichloramine (NHCl₂) as follows:



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After it is formed, dichloramine readily enters the gas phase, where it may be inhaled. In the lungs, dichloramine reacts with water to produce hydrochloric acid and ammonia, as well as reactive oxygen species. In small doses, these products cause mild irritation, but inhalation of large amounts of chloramines can lead to pneumonitis (inflammation of the lungs) and may require medical intervention.

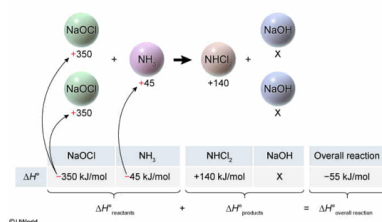
Based on the information given in the passage, which of the following is the standard molar enthalpy of formation for sodium hydroxide?

- ☐ A. -590 kJ/mol [22%]
☒ B. -470 kJ/mol [42%]
☐ C. -940 kJ/mol [6%]
☐ D. 275 kJ/mol [28%]

Correct

42% answered correctly

Explanation:



According to **Hess's law**, the overall **enthalpy change ΔH°** of a reaction is equal to the **sum of changes** in enthalpy of each component. In the passage, Table 1 gives the standard molar enthalpies of formation of each component *except* sodium hydroxide (NaOH). Therefore, the enthalpy of formation for NaOH can be calculated by subtracting the enthalpies of bleach, ammonia, and dichloramine from the overall enthalpy of the reaction (-55 kJ/mol) and applying the following:

- **The signs of the reactants must be changed.** Table 1 gives enthalpies of formation, but ammonia and bleach are consumed, not formed. Therefore, they contribute $+350 \text{ kJ/mol}$ and $+45 \text{ kJ/mol}$ of enthalpy, respectively.
- The **standard enthalpies of formation** for each component must be **multiplied** by their **stoichiometric coefficients**. In the case of bleach, two molecules are consumed per reaction, so the standard enthalpy must be multiplied by 2. Similarly, two molecules of sodium hydroxide are formed per reaction, and these must be accounted for.

To determine the standard enthalpy of formation of NaOH (x), set up the equation:

$$2(350 \text{ kJ/mol}) + 45 \text{ kJ/mol} + 140 \text{ kJ/mol} + 2x = -55 \text{ kJ/mol}$$

Solve for the enthalpy of formation of NaOH (x).

$$2x = -55 \text{ kJ/mol} - 140 \text{ kJ/mol} - 45 \text{ kJ/mol} - 700 \text{ kJ/mol} = -940 \text{ kJ/mol}$$

$$x = -470 \text{ kJ/mol}$$

Alternatively, the answer can be found without directly changing the signs of the given enthalpies by using Hess's law and stating the overall enthalpy as the difference between the sums of the product and reactant enthalpies.

$$\Delta H_{\text{overall reaction}}^\circ = \Delta H_{\text{products}}^\circ - \Delta H_{\text{reactants}}^\circ$$

$$\Delta H_{\text{overall reaction}}^{\circ} = \left(\Delta H_{\text{NHCl}_2}^{\circ} + \Delta H_{\text{NaOH}}^{\circ} \right) - \left(\Delta H_{\text{NaOCl}}^{\circ} + \Delta H_{\text{NH}_3}^{\circ} \right)$$

$$- 55 \text{ kJ/mol} = (140 \text{ kJ/mol} + 2x) - [2(- 350 \text{ kJ/mol}) + (- 45 \text{ kJ/mol})]$$

$$x = - 470 \text{ kJ/mol}$$

(Choice A) Using the enthalpies of formation for 1 mol of bleach and NaOH instead of multiplying by 2 yields this answer.

(Choice C) Solving for 2x rather than x yields this answer, which is the enthalpy of formation for 2 mol of NaOH.

(Choice D) Keeping the signs of the enthalpies for bleach and ammonia negative instead of switching them to positive yields this answer.

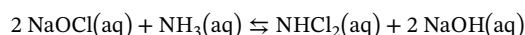
Educational objective:

Hess's law states that the overall enthalpy change of a reaction is equal to the sum of the enthalpies of formation for each component. Because reactants are consumed rather than formed, the signs of their enthalpies must be changed. Each component must be multiplied by its stoichiometric coefficient to account for the total number of molecules consumed and formed in each reaction.

Time spent: 0 sec
Subject:
 General Chemistry

QID: 401189
System:
 Thermodynamics, Kinetics & Gas Laws

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After it is formed, dichloramine readily enters the gas phase, where it may be inhaled. In the lungs, dichloramine reacts with water to produce hydrochloric acid and ammonia, as well as reactive oxygen species. In small doses, these products cause mild irritation, but inhalation of large amounts of chloramines can lead to pneumonitis (inflammation of the lungs) and may require medical intervention.

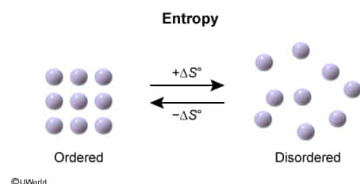
During the formation of dichloramine and NaOH, the order (organization) of the system:

- ☐ A. decreases because entropy is always increasing. [13%]
- ☐ B. stays the same because the number of moles stays the same. [18%]
- ☒ C. increases because the change in entropy is negative. [62%]
- ☐ D. does not change because the system is at equilibrium. [5%]

Correct

63% answered correctly

Explanation:



Entropy is a measure of the **disorder or randomness** of a system. For example, a crystal has a high degree of order and therefore has low entropy because the atoms that compose it are aligned and spaced evenly. On the other hand, a gas is disordered and has

high entropy because its individual molecules are randomly spaced and oriented. The overall entropy of a system changes during a reaction, and the **change in entropy** is expressed by ΔS° . When ΔS° for a reaction is **positive**, it indicates that the system is becoming **more random** (less ordered) whereas a **negative ΔS°** corresponds to **higher order**.

The passage gives a negative value for ΔS° of the overall reaction ($-150 \text{ J/mol}\cdot\text{K}$). Therefore, over the course of the reaction, the entropy of the system *decreases* and the order *increases*.

(Choice A) The second law of thermodynamics states that the entropy (disorder) of the universe is always increasing. However, this law applies only to the universe as a whole, and individual systems may have decreasing entropy as expressed by a negative ΔS° .

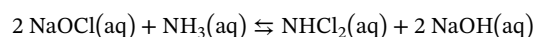
(Choice B) Although a change in entropy is often accompanied by a change in the number of moles, many other factors also contribute to entropy. The disorder of a system must be measured empirically by ΔS° .

(Choice D) A system at equilibrium is not undergoing any net reactions, and therefore does not undergo a change in entropy. However, because products are being formed, the system described in the passage is not at equilibrium.

Educational objective:

The change in entropy (disorder) of a system over the course of a reaction is expressed by ΔS° and must be measured empirically. A positive ΔS° corresponds to an increase in entropy (more disordered) whereas a negative ΔS° corresponds to a decrease in entropy (more ordered).

Household cleaners commonly contain either ammonia (NH₃) or bleach (NaOCl) as the principal active ingredient. Mixing ammonia-based and bleach-based cleaners can be hazardous, as the resulting reactions can form several toxic products. One of these reactions forms the irritant dichloramine (NHCl₂) as follows:



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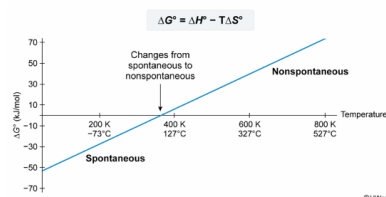
Assuming ΔH° and ΔS° are constant at all temperatures, at approximately which temperature (in degrees Celsius) will ΔG° of Reaction 1 equal zero (ie, switch from spontaneous to nonspontaneous)?

- ✓ ☒ A. 95 °C [35%]
☐ B. 370 °C [28%]
☐ C. -270 °C [31%]
☐ D. -640 °C [5%]

Correct

34% answered correctly

Explanation:



The change in **Gibbs free energy** (ΔG°) of a reaction under standard conditions (ie, all reactants and products are initially at 1 M concentration) determines whether the reaction is **spontaneous** ($\Delta G^\circ < 0$) or **nonspontaneous** ($\Delta G^\circ > 0$). The ΔG° of the reaction depends on the **change in enthalpy** ΔH° , the **change in entropy** ΔS° , and temperature T (in **Kelvin**) of the system according to the following **equation**:

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

The point at which a reaction transitions from being spontaneous to nonspontaneous (or vice versa) depends on the system temperature and occurs when $\Delta G^\circ = 0$. As such, the limit of favorability and the transition temperature of a reaction can be calculated by setting $\Delta G^\circ = 0$ and substituting the given values of ΔH° and ΔS° for the overall reaction into the equation.

$$0 = \Delta H^\circ - T\Delta S^\circ = -55 \frac{\text{kJ}}{\text{mol}} - T\left(-150 \frac{\text{J}}{\text{mol} \cdot \text{K}}\right)$$

Converting the units of ΔS° to kJ/mol·K yields

$$0 = -55 \frac{\text{kJ}}{\text{mol}} - T\left(-150 \frac{\text{J}}{\text{mol} \cdot \text{K}} \times \frac{1 \text{ kJ}}{1,000 \text{ J}}\right) = -55 \frac{\text{kJ}}{\text{mol}} - T\left(-0.15 \frac{\text{kJ}}{\text{mol} \cdot \text{K}}\right)$$

Solving for temperature gives

$$T = \frac{(-55 \text{ kJ} \cdot \text{mol}^{-1})}{(-0.15 \text{ kJ} \cdot \text{K}^{-1} \cdot \text{mol}^{-1})} = 367 \text{ K}$$

Converting the temperature in kelvin to Celsius yields

$$367 - 273 = 94\text{ }^{\circ}\text{C} \approx 95\text{ }^{\circ}\text{C}$$

(Choice B) Keeping the temperature in kelvin yields an answer of approximately 370 K, not 370 °C.

(Choice C) Keeping the entropy in J/mol·K instead of converting to kJ/mol·K yields an answer of approximately -270 °C.

(Choice D) Dropping a negative sign anywhere during the calculation will lead to an answer of approximately -640 °C. Note that absolute zero is approximately -273 °C, so -640 °C is not physically possible.

Educational objective:

The Gibbs free energy of a reaction determines whether the reaction is spontaneous. It is related to both enthalpy and entropy by the equation $\Delta G^{\circ} = \Delta H^{\circ} - T\Delta S^{\circ}$. A negative ΔG° corresponds to a spontaneous reaction whereas a positive ΔG° corresponds to a nonspontaneous reaction.

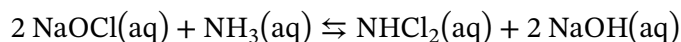
Time spent: 0 sec

Subject:
General Chemistry

QID: 401191

System:
Thermodynamics, Kinetics & Gas Laws

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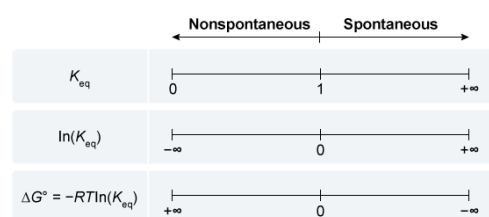
If dichloramine formation is spontaneous under standard conditions, the equilibrium constant must be:

- ☐ A. < 0. [9%]
☐ B. > 0 and < 1. [9%]
☐ C. = 1. [2%]
☒ D. > 1. [77%]

Correct

78% answered correctly

Explanation:



A reaction is **spontaneous** in the **standard state** (all reactants and products initially at 1 M concentration) if ΔG° is **negative**. By definition,

$$\Delta G^\circ = -RT \ln(K_{eq})$$

where R is the universal gas constant, T is the temperature in kelvin, and K_{eq} is the **equilibrium constant** of the reaction at the given temperature. In the standard state, K_{eq} is related to spontaneity:

- If $\ln(K_{\text{eq}})$ is **positive**, ΔG° is negative and the **reaction is spontaneous**. The natural logarithm ($\ln$) of any number greater than 1 is positive, so K_{eq} must be greater than 1 for the reaction to be spontaneous under standard state conditions.
- If $\ln(K_{\text{eq}})$ is **equal to 0**, the entire right side of the equation is equal to 0 and the system is at **equilibrium** in the standard state. This occurs when $K_{\text{eq}} = 1$ because $\ln(1) = 0$ (**Choice C**).
- If $\ln(K_{\text{eq}})$ is **negative**, then a negative number ($-RT$) is multiplied by another negative number ($\ln[K_{\text{eq}}]$), making ΔG° positive and the reaction **nonspontaneous**. The natural logarithm of any number between 0 and 1 is a negative number (**Choice B**).

(Choice A) The natural logarithm ($\ln$) of zero or a negative number is undefined. Therefore, it is impossible for the equilibrium constant of any reaction to be less than or equal to zero.

Educational objective:

For a reaction to be spontaneous in the standard state, ΔG° must be negative. Because $\Delta G^\circ = -RT \ln(K_{\text{eq}})$ and the natural logarithm of any number greater than 1 is positive, K_{eq} must be greater than 1 for a standard state reaction to be spontaneous. Similarly, a K_{eq} between 0 and 1 corresponds to nonspontaneous reactions, and a K_{eq} equal to 1 corresponds to a reaction at equilibrium in the standard state.

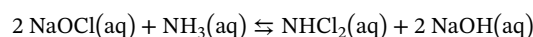
Time spent: 0 sec

Subject:
General Chemistry

QID: 401192

System:
Thermodynamics, Kinetics & Gas Laws

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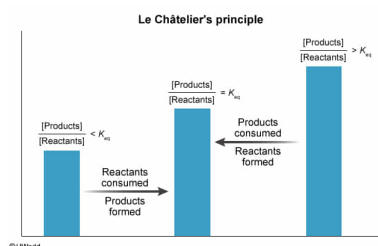
If large amounts of ammonia and bleach were mixed, emergency responders could help stop the reaction by doing which of the following?

- ✓ ☒ A. Mixing sodium hydroxide into the solution [85%]
☐ B. Adding ammonia to the solution [2%]
☐ C. Removing dichloramine from the solution [6%]
☐ D. Evaporating water from the solution [5%]

Correct

84% answered correctly

Explanation:



A reaction's **equilibrium state** is the set of concentrations of products and reactants at which **no new products or reactants** are formed. Under equilibrium conditions, the ratio of product concentrations to reactant concentrations equals the equilibrium constant for the reaction, K_{eq} . Note that this is not necessarily the same as when product concentration equals reactant concentration, in which case K_{eq} would equal 1.

Le Châtelier's principle states that any change to a system that moves it away from equilibrium, such as the addition of products or reactants, will cause it to **adjust** accordingly to **come back to equilibrium**. Therefore, the **addition of reactants** to the system will drive the formation of **more products**, and the **addition of products** will drive the **formation of reactants**. Similarly, the removal of reactants drives the system toward reactant formation, and the removal of products drives the system toward product formation.

Sodium hydroxide (NaOH) is one of the products of the reaction described in the passage. Therefore, the addition of NaOH will drive the reaction toward reactant formation and away from product formation, which will help stop the reaction from continuing.

(Choice B) Ammonia is one of the reactants. Adding it to the solution will drive the formation of more products.

(Choice C) Dichloramine is one of the products of the reaction, and its removal from the system will drive further product formation. During the reaction, dichloramine naturally leaves the solution and enters the gas phase, and the reaction is induced to form *more* dichloramine.

(Choice D) Evaporating (removing) water from the reaction would concentrate its components. Increasing the concentration of the reactants and products causes the reaction to shift in the direction that produces the fewest number of ions. Because the number of reactant ions ($2 \text{NaOCl(aq)} + 1 \text{NH}_3(\text{aq}) = 3 \text{ ions total}$) is the same as the number of product ions ($1 \text{NHCl}_2(\text{aq}) + 2 \text{NaOH(aq)} = 3 \text{ ions total}$), the equilibrium position is unaffected.

Educational objective:

Le Châtelier's principle states that changes to a system that move it away from equilibrium will cause the system to adjust and come back to equilibrium. The addition of reactants or removal of products results in product formation, and the addition of products or removal of reactants results in reactant formation.

Time spent: 0 sec

Subject:
General Chemistry

QID: 401193

System:
Thermodynamics, Kinetics & Gas Laws